

REVIEW of MATHEMATICAL INDUCTION

PRINCIPAL of mathematical induction

Given a proposition function $P(n)$ defined for integers n , and a fixed integer a

Then, if these two conditions are true

1 $P(a)$ is true

2 if $P(k)$ is true for some integer $k \geq a$, then $P(k+1)$ is also true

Then $P(n)$ is true for all integers $n \geq a$

OUTLINE for Mathematical Induction

To show that a propositional function $P(n)$ is true for all integers $n \geq a$, follow these steps

BASE STEP: Verify that $P(a)$ is true

INDUCTIVE STEP: Show that if $P(k)$ is true for some integer $k \geq a$, then $P(k+1)$ is also true

• Assume $P(k)$ is true for an arbitrary integer, k with $k \geq a$ (this is **INDUCTIVE HYPOTHESIS**)

• With this assumption (the inductive hypothesis), show $P(k+1)$ is true

PROOF by mathematical induction

→ Many results in mathematics are claimed true for every positive integer

→ Any of these results could be checked for a specific value of n (e.g. 1, 2, 3, ...)
but it would be impossible to check every possible case

PROOF by Induction

→ a mathematical technique used to prove statements about an infinite set of objects, such as all set of natural numbers

PROOF by Induction in Computer Science

→ a mathematical technique used to prove statements when dealing with recursively defined objects such as trees, expressions and more

Example Let S_n represent the statement that the sum of the first n positive integers

$$\text{is } \frac{n+1}{2}$$

$$S_n: 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

if $n = 1$, then S_1 is $1 = \frac{1(1+1)}{2}$, which is TRUE.

if $n = 2$, then S_2 is $1 + 2 = \frac{2(2+1)}{2}$, which is TRUE

if $n = 3$, then S_3 is $1 + 2 + 3 = \frac{3(3+1)}{2}$, which is TRUE

if $n = 4$, then S_4 is $1 + 2 + 3 + 4 = \frac{4(4+1)}{2}$, which is TRUE

Continuing in this way for any amount of time would still not prove that S_n is true for every positive integer value of n

To prove that such statements are true

This Principle is often used

Base Case

Assumption

ODD: $2k+1$

EVEN: $2k$

POSITIVE numbers $k, k \in \mathbb{Z}^+$

NEGATIVE numbers $k, k \in \mathbb{Z}^-$

1. Let S_n be a statement concerning the positive

integer n . Suppose that

1. S_1 is true

2. for any positive integer k , $k \leq n$, if S_k is true,

then S_{k+1} is also true

2. Then S_n is true for every positive integer value of n

$$S_n = \frac{n(n+1)}{2}$$

S_k is true $\rightarrow$ prove true

$$S_k = \frac{k(k+1)}{2} \rightarrow \text{but what about this?}$$

if k is true then $k+1$ should also be true

$$\frac{k(k+1)}{2} + \frac{k+1}{1} = \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

This is the proper way of solving this type of problem

PROVE $S(n)$ is sum of all positive integers including n

$$S(n) = \frac{n(n+1)}{2}$$

Base Case = prove using $n=1$

Induction Step = assume $S(k)$ is true, where k is any positive integer

then $S(k+1)$ must be true

Base Case:

$$S(1) = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$$

Prove

$$S(n) = \frac{n(n+1)}{2}$$

$$S(1) = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$$

This is TRUE

Induction Case:

$$\text{Assume } S(k) = \frac{k(k+1)}{2} \text{ where } k, k \in \mathbb{Z}^+$$

$$\text{Assume } S(k) = \frac{k(k+1)}{2}$$

If $S(k)$ is true then $S(k+1)$ should also be true

$$\sum_{i=1}^k = \frac{k(k+1)}{2}$$

$$\sum_{i=1}^{k+1} = \frac{k(k+1)}{2} + k+1$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$\sum_{i=1}^{k+1} = \frac{(k+1) + (k+2)}{2}$$

My Answer

$\therefore$ Hence, $S(n)$ the sum of all positive integers is true.

How it should be

$S(n)$ is the sum of all positive integers

$$S_n = \frac{n(n+1)}{2}$$

Base Case: $n=1$

$$S_1 = \frac{1(1+1)}{2} = 1$$

This is TRUE

Induction Case:

$$\text{Assuming } S(k) = \frac{k(k+1)}{2} \text{ is true}$$

If k is true then $k+1$ should also be true

$$\sum_{i=1}^k = \frac{k(k+1)}{2} \leftarrow \text{Induction hypothesis}$$

Generalized Principle of Mathematical Induction

$\rightarrow$ some statements S_n are not true for the first few values

of n , but are true for all values of n that are

greater than or equal to some fixed integer j .

So, Induction has 3 main steps

1. BASE CASE = 1st thing is TRUE

2. INDUCTIVE HYPOTHESIS = Assume is true $n \leq k$

$$\begin{aligned}
 \sum_{i=1}^{k+1} i &= \sum_{i=1}^k i + k+1 \\
 &= \frac{k(k+1)}{2} + k+1 \\
 &= \frac{k(k+1) + 2(k+1)}{2} \\
 &= \frac{(k+1) + (k+2)}{2} \\
 \text{Thus, } \sum_{i=1}^{k+1} i &= \frac{(k+1) + (k+2)}{2}
 \end{aligned}$$

Show that $k+1$ is true

3. CONCLUSION

= Every n is True

Youtube Tutorial

Example 1 Show $1+2+\dots+n = \frac{n(n+1)}{2}$

Base case: $n=1$

$$S_n = \frac{n(n+1)}{2}$$

$$S(1) = \frac{1(1+1)}{2}$$

$$S(1) = 1$$

which is True

Inductive Hypothesis

Assuming $S(k) = \frac{k(k+1)}{2}$ is True or $n \leq k$ is True

$$\sum_{i=1}^{k+1} i = \frac{k(k+1)}{2}$$

if k is True then $k+1$ is also true

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^k i + (k+1)$$

$$= \frac{k(k+1)}{2} + \frac{k+1}{1}$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$$

Conclusion Thus $S(n) = \frac{n(n+1)}{2}$ is True

Example: Show $\overline{(A_1 \cup A_2 \cup \dots \cup A_n)} = \bar{A}_1 \cap \bar{A}_2 \cap \dots \bar{A}_n$

Base Case $\overline{A_1 \cup A_2} = \bar{A}_1 \cap \bar{A}_2$

According to DeMorgan's this is True

Inductive Hypothesis:

$$\overline{A_1 \cup \dots \cup A_k} = \bar{A}_1 \cap \dots \cap \bar{A}_k$$

Assuming $\overline{A_1 \cup \dots \cup A_k}$ true.

$$\overline{(A_1 \cup \dots \cup A_k) \cup A_{k+1}}$$

Using De Morgan's Law

$$\overline{(A_1 \cup \dots \cup A_k) \cup A_{k+1}} = (\bar{A}_1 \cap \dots \cap \bar{A}_k) \cap \bar{A}_{k+1} \text{ is True}$$

Conclusion: Thus, $\overline{(A_1 \cup A_2 \cup \dots \cup A_n)} = \bar{A}_1 \cap \bar{A}_2 \cap \dots \bar{A}_n$ is True

RECAP of laws of Algebras

1. IDEMPOTENT LAWS

$$A \cup A = A$$

$$A \cap A = A$$

2. ASSOCIATIVE LAWS

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

3. COMMUTATIVE LAWS

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

4. DISTRIBUTIVE LAWS

5. IDENTITY LAWS

$$A \cup \emptyset = A$$

$$A \cap U = A$$

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

6. INVOLUTION LAWS

$$(A^c)^c = A$$

7. COMPLEMENT LAWS

$$A \cup A^c = U \text{ or } A \cup \bar{A} = U$$

8. DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c \text{ or } \overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$(A \cap B)^c = A^c \cup B^c \text{ or } \overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cap A^c = \emptyset \quad \text{or} \quad A \cap \bar{A} = \emptyset$$

$$U^c = \emptyset \quad \text{or} \quad \bar{U} = \emptyset$$

$$\emptyset^c = U \quad \text{or} \quad \overline{\emptyset} = U$$

PROVING

1. Idempotent law #1a Proof

$$A \cup A = A$$

$$A \cap A = A$$

$$A \cup A = \{x \mid x \in A \vee x \in A\} \quad A \cap A = \{x \mid x \in A \wedge x \in A\}$$

$$A \cup A = \{x \mid x \in A\} \quad A \cap A = \{x \mid x \in A\}$$

$$A \cup A = A \quad A \cap A = A$$

2. ASSOCIATIVE LAWS

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = \{x \mid x \in A \wedge (x \in B \wedge x \in C)\}$$

$$A \cup (B \cup C) = \{x \mid x \in A \vee (x \in B \vee x \in C)\}$$

Base from associative law of logical equivalences
 $p \wedge (q \wedge r) = (p \wedge q) \wedge r$

Base from associative law of logical equivalences
 $p \vee (q \vee r) = (p \vee q) \vee r$

$$A \cap (B \cap C) = \{x \mid (x \in A \wedge x \in B) \wedge x \in C\}$$

$$A \cup (B \cup C) = \{x \mid (x \in A \vee x \in B) \vee x \in C\}$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

3. COMMUTATIVE LAW

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

$$A \cup B = \{x \mid x \in A \vee x \in B\} \quad A \cap B = \{x \mid x \in A \wedge x \in B\}$$

$$A \cup B = \{x \mid x \in B \vee x \in A\} \quad A \cap B = \{x \mid x \in B \wedge x \in A\}$$

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

4. DISTRIBUTIVE LAWS

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = \{x \mid x \in A \wedge (x \in B \vee x \in C)\}$$

$$A \cup (B \cap C) = \{x \mid x \in A \vee (x \in B \wedge x \in C)\}$$

$$A \cap (B \cup C) = \{x \mid x \in A \wedge (x \in B \vee x \in C)\}$$

Base from rules of logic
 $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\}$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

5. IDENTITY LAWS

$$A \cup \emptyset = A$$

$$A \cap U = A$$

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

$$A \cup \emptyset = \{x \mid x \in A \vee x \in \emptyset\}$$

$$A \cap U = \{x \mid x \in A \wedge x \in U\}$$

$$A \cup U = \{x \mid x \in A \vee x \in U\}$$

$$A \cap \emptyset = \{x \mid x \in A \wedge x \in \emptyset\}$$

Note $x \in \emptyset$ is False

Note $x \in U$ is True

$$A \cup U = \{x \mid x \in A \vee T\}$$

$$A \cap \emptyset = \{x \mid x \in A \wedge F\}$$

$$A \cup \emptyset = \{x \mid x \in A \vee F\}$$

$$A \cap U = \{x \mid x \in A \wedge T\}$$

$$A \cup U = \{x \mid T\}$$

$$A \cap \emptyset = \{x \mid F\}$$

$$A \cup \emptyset = \{x \mid x \in A\}$$

$$A \cap U = \{x \mid x \in A\}$$

Note: $x \in U = T$

Note $x \in \emptyset = F$

$$A \cup \emptyset = A$$

$$A \cap U = A$$

$$A \cup U = \{x \mid x \in U\}$$

$$A \cap \emptyset = \{x \mid x \in \emptyset\}$$

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

6. INVOLUTION LAW

$$(A^c)^c = A$$

$$(A^c)^c = \{x \mid x \in A^c\}$$

$$(A^c)^c = \{x \mid \neg(x \in A^c)\}$$

$$(A^c)^c = \{x \mid \neg(x \notin A)\}$$

$$(A^c)^c = \{x \mid \neg(\neg(x \in A))\}$$

$$(A^c)^c = \{x \mid (x \in A)\}$$

$$(A^c)^c = A$$

7. DE MORGAN'S LAW

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cup B)^c = \{x \mid x \notin (A \cup B)\}$$

$$(A \cup B)^c = \{x \mid \neg(x \in (A \cup B))\}$$

$$(A \cup B)^c = \{x \mid \neg(x \in A \vee x \in B)\}$$

$$(A \cup B)^c = \{x \mid (x \notin A \wedge x \notin B)\}$$

$$(A \cup B)^c = \{x \mid (x \in A^c \wedge x \in B^c)\}$$

$$(A \cup B)^c = \{x \mid (x \in (A^c \cap B^c))\}$$

$$(A \cup B)^c = \{x \mid (x \in (A^c \cap B^c))\}$$

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cap B)^c = \{x \mid x \notin (A \cap B)\}$$

$$(A \cap B)^c = \{x \mid \neg(x \in (A \cap B))\}$$

$$(A \cap B)^c = \{x \mid \neg(x \in A \wedge x \in B)\}$$

$$(A \cap B)^c = \{x \mid (x \notin A \vee x \notin B)\}$$

$$(A \cap B)^c = \{x \mid (x \in A^c \vee x \in B^c)\}$$

$$(A \cap B)^c = \{x \mid (x \in (A^c \cup B^c))\}$$

$$(A \cap B)^c = A^c \cup B^c$$

